BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 042 · Programming Techniques · Value and Reference

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a local variable?

Q2. Why use a method?

Q3. What does an accumulator store?

Q4. What is a hash collision?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — by value

Recall the term — ref

Recall the term — reference

Recall the term — argument

Recall the term — mutation

New learning

Explain the important decision — C# int values are passed by value by default; a method changing its parameter does not change the caller's int.

Worked example

Compare changes to a copy and an object

1. Identify the task — Compare changes to a copy and an object

2. Follow the logic — C# int values are passed by value by default; a method changing its parameter does not change the caller's int.

3. Check the outcome — Explain how each step contributes to the result. Test an edge case.

We Do · Practise with guidance

Problem 1

If a method takes ref int n and adds 2, starting 3, what is the caller value?

Discuss, then respond

Check your thinking

5.

Problem 2

Does passing a reference-type object by value copy the object itself?

No; the reference is copied.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Explain why changing a normal int parameter does not change the original. [2]

Question 2

Apply independently
Write a C# method header to increment an integer by reference. [2]

Question 3

Apply independently
Distinguish passing an object reference by value from using ref. [3]

Exit Ticket · Answer before checking

True or false: a reference-type argument passed normally allows the method to replace the caller's variable.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A name scoped to a function or block

Q2. To modularise and reuse logic

Q3. A running total

Q4. Two keys produce the same location

You Do pages · Independent questions

Q1. The method receives a copy of its value.

Q2. static void Inc(ref int value)

Q3. The former copies the reference, permitting object mutation; ref permits reassignment of the caller's variable.

Exit Ticket · Model answer

False. Passing a reference-type variable normally passes a copy of the reference. Assigning a different object to that parameter does not reassign the caller's variable.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: